

Telco Customer Churn & Service Analysis

1. Project Overview

A telecommunications company is experiencing customer churn but lacks a clear understanding of the customer, service and lifecycle characteristics associated with customers leaving the business.

Management needs a clear view of who is churning, how their service engagement differs and at what stage of the customer lifecycle churn is most concentrated.

This project analyzes customer-level telecommunications data to identify observable churn patterns across customer characteristics, service engagement, tenure and contract structure.

Business Questions

The analysis focuses on three business areas:

1. Which customer characteristics are associated with higher churn?
2. How is customer service engagement associated with churn?
3. At what stage of the customer lifecycle is churn most concentrated, and how does contract type relate to this pattern?

2. Dataset & Methodology

2.1 Source Data

The project began with a set of telecommunications datasets covering different aspects of the customer base including demographics, services, customer status, location and population information.

Dataset	Rows	Columns	Purpose
Demographics	7,043	9	Customer demographic characteristics
Services	7,043	31	Service usage, tenure, contracts and billing information
Status	7,043	12	Customer status and churn information
Location	7,043	10	Customer geographic information
Population	1,671	3	Geographic population information

2.2 Tools & Technologies

The analysis was completed using:

- **Python** – Data Exploration, Cleaning and Validation
- **PostgreSQL** – Storage of the cleaned customer-level dataset and SQL-based business analysis
- **Power BI** – Interactive Dashboard and Visualization

2.3 Analytical Approach

The project followed an end-to-end analytical workflow:

Source Datasets → Data Exploration → Data Cleaning & Validation → Cleaned Customer-Level Dataset → PostgreSQL → Derived Analytical Fields → SQL Business Analysis → Power BI Dashboard → Findings & Recommendations

The analysis was organized into three main areas:

Customer Characteristics

- Age group
- Marital and dependent status
- Referral engagement

Service Engagement

- Number of subscribed services
- Internet service type

Customer Lifecycle & Contract

- Tenure stage
- Contract type
- Contract type across different tenure stages

For each segment, the analysis calculates observed churn rate:

Churn Rate = Churned Customers / Total Customers in the Group

The analysis is descriptive and observational. Therefore, differences in churn rates are interpreted as associations and patterns in the dataset rather than evidence of causation.

3. Data Preparation & Transformation

3.1 Data Cleaning

1. Demographics Table

- Standardized all column headers to snake_case.
- Removed redundant count column.

2. Services Table

- Standardized all column headers to snake_case.
- Replaced missing Internet Type with 'No Internet Service' for customers without internet service.
- Removed unnecessary columns: service_id, count and quarter.

3. Status Table

- Standardized all columns headers into snake_case.
- Filled missing values in both churn_category and churn_reason with 'Not Churned' for customers who had not churned.
- Dropped status_id, count, quarter, churn_value, churn_score and cltv to keep only the primary final churn outcomes and satisfaction scores.

4. Location Table

- Standardized all column headers to snake_case.
- Dropped location_id, count, country, state and lat_long.

5. Population Table

- Standardized all the column headers to snake_case.
- Stripped comma separators out of the population text strings and converted the column into integer.
- Dropped the arbitrary and redundant id column.

3.2 Data Integration

The five source datasets were combined using customer_id and zip_code to create a single customer-level dataset for analysis.

3.3 Derived Analytical Fields

Three analytical fields were created to support the business questions and make the analysis easier to interpret.

Age Group

Customers were grouped into:

- 18-29
- 30-44
- 45-59
- 60+

The grouping provides a compact way to compare observed churn across different age segments.

Service Count

A service_count field was created to represent the number of applicable services subscribed to by each customer. The count ranges from 0 to 9, based on the service categories included in the calculation.

Internet service type was not counted separately because it describes the types of internet service rather than representing an additional service category.

Tenure Group

Customers were grouped according to their tenure:

- 0-12 Months
- 13-24 Months
- 25-48 Months
- 49-60 Months
- 61+ Months

These groups allow churn to be examined across different stages of the customer lifecycle.

4. Analysis & Key Findings

4.1 Customer Characteristics

Analysis:

Customer churn was examined across age group, marital and dependent status and referral engagement.

Age Group

	age_group character varying (20) 🔒	total_customers bigint 🔒	churned_customers bigint 🔒	retained_customers bigint 🔒	churn_rate_percentage numeric 🔒
1	18-29	1401	304	1097	21.70
2	30-44	1943	475	1468	24.45
3	45-59	1917	459	1458	23.94
4	60+	1782	631	1151	35.41

Key Finding:

Customers aged 60+ had the highest observed churn rate (35.41%) while customers aged 18-29 recorded 21.70%.

Marital & Dependent Status

	married character varying (10)	dependents character varying (10) 🔒	total_customers bigint 🔒	churned_customers bigint 🔒	retained_customers bigint 🔒	churn_rate_percentage numeric 🔒
1	No	No	3339	1150	2189	34.44
2	Yes	No	2077	613	1464	29.51
3	No	Yes	302	50	252	16.56
4	Yes	Yes	1325	56	1269	4.23

Key Finding:

Customers who were married with dependents had the lowest churn (4.2%) while those who were neither married nor had dependents had 34.4% churn.

Referral Engagement

	referral_group text	total_customers bigint	churned_customers bigint	retained_customers bigint	churn_rate_percentage numeric
1	0 Referrals	3821	1245	2576	32.58
2	1-2 Referrals	1322	533	789	40.32
3	3-5 Referrals	755	71	684	9.40
4	6+ Referrals	1145	20	1125	1.75

Key Finding:

Churn was 40.3% for customers with 1-2 referrals, but only 1.8% for customers with 6+ referrals, showing a non-linear relationship.

4.2 Service Engagement

Analysis:

Churn was examined across the number of subscribed services and internet service types.

Number of Services

Churn varied considerably by the number of subscribed services.

	service_count integer	total_customers bigint	churned_customers bigint	retained_customers bigint	churn_rate_percentage numeric
1	0	73	32	41	43.84
2	1	1668	329	1339	19.72
3	2	1081	340	741	31.45
4	3	811	291	520	35.88
5	4	786	273	513	34.73
6	5	787	246	541	31.26
7	6	746	187	559	25.07
8	7	558	121	437	21.68
9	8	348	41	307	11.78
10	9	185	9	176	4.86

Key Finding:

Churn varied non-linearly, ranging from 43.8% for customers with 0 services to 4.9% for customers with 9 services. The 0-service group contained only 73 customers.

Internet Service Type

	internet_type character varying (30) 🔒	total_customers bigint 🔒	churned_customers bigint 🔒	retained_customers bigint 🔒	churn_rate_percentage numeric 🔒
1	Fiber Optic	3035	1236	1799	40.72
2	Cable	830	213	617	25.66
3	DSL	1652	307	1345	18.58
4	No Internet Service	1526	113	1413	7.40

Key Finding:

Fiber Optic customers had the highest observed churn rate (40.72%), followed by Cable (25.66%), DSL (18.58%) and No Internet Service (7.40%).

4.3 Customer Lifecycle & Contract

Analysis:

Churn was examined across tenure stages, contract types and the relationship between contract type and tenure.

Tenure Stage

	tenure_group character varying (20) 🔒	total_customers bigint 🔒	churned_customers bigint 🔒	retained_customers bigint 🔒	churn_rate_percentage numeric 🔒
1	0-12 Months	2186	1037	1149	47.44
2	13-24 Months	1024	294	730	28.71
3	25-48 Months	1594	325	1269	20.39
4	49-60 Months	832	120	712	14.42
5	61+ Months	1407	93	1314	6.61

Key Findings:

Churn was highest during 0-12 months (47.44%) and declined at 6.6% for 61+ months.

This indicates that churn is strongly concentrated in the early customer lifecycle.

Contract Type

	contract character varying (30) 🔒	total_customers bigint 🔒	churned_customers bigint 🔒	retained_customers bigint 🔒	churn_rate_percentage numeric 🔒
1	Month-to-Month	3610	1655	1955	45.84
2	One Year	1550	166	1384	10.71
3	Two Year	1883	48	1835	2.55

Key Finding:

Month-to-month customers had 45.84% churn, compared with 10.71% for one-year contracts and 2.55% for two-year contracts.

Contract Type Across Tenure Stages

The interaction between contract type and tenure provides additional context.

	tenure_group character varying (20) 🔒	contract character varying (30) 🔒	total_customers bigint 🔒	churned_customers bigint 🔒	retained_customers bigint 🔒	churn_rate_percentage numeric 🔒
1	0-12 Months	Month-to-Month	1913	1024	889	53.53
2	0-12 Months	One Year	146	13	133	8.90
3	0-12 Months	Two Year	127	0	127	0.00
4	13-24 Months	Month-to-Month	686	278	408	40.52
5	13-24 Months	One Year	215	16	199	7.44
6	13-24 Months	Two Year	123	0	123	0.00
7	25-48 Months	Month-to-Month	715	264	451	36.92
8	25-48 Months	One Year	546	55	491	10.07
9	25-48 Months	Two Year	333	6	327	1.80
10	49-60 Months	Month-to-Month	209	65	144	31.10
11	49-60 Months	One Year	326	44	282	13.50
12	49-60 Months	Two Year	297	11	286	3.70
13	61+ Months	Month-to-Month	87	24	63	27.59
14	61+ Months	One Year	317	38	279	11.99
15	61+ Months	Two Year	1003	31	972	3.09

Key Finding:

Month-to-month customers had the highest churn rate at every tenure stage, reaching 53.53% during the first 12 months.

4.4 Power BI Dashboard

The dashboard visualizes the main churn patterns using customer, tenure, contract and age segments.

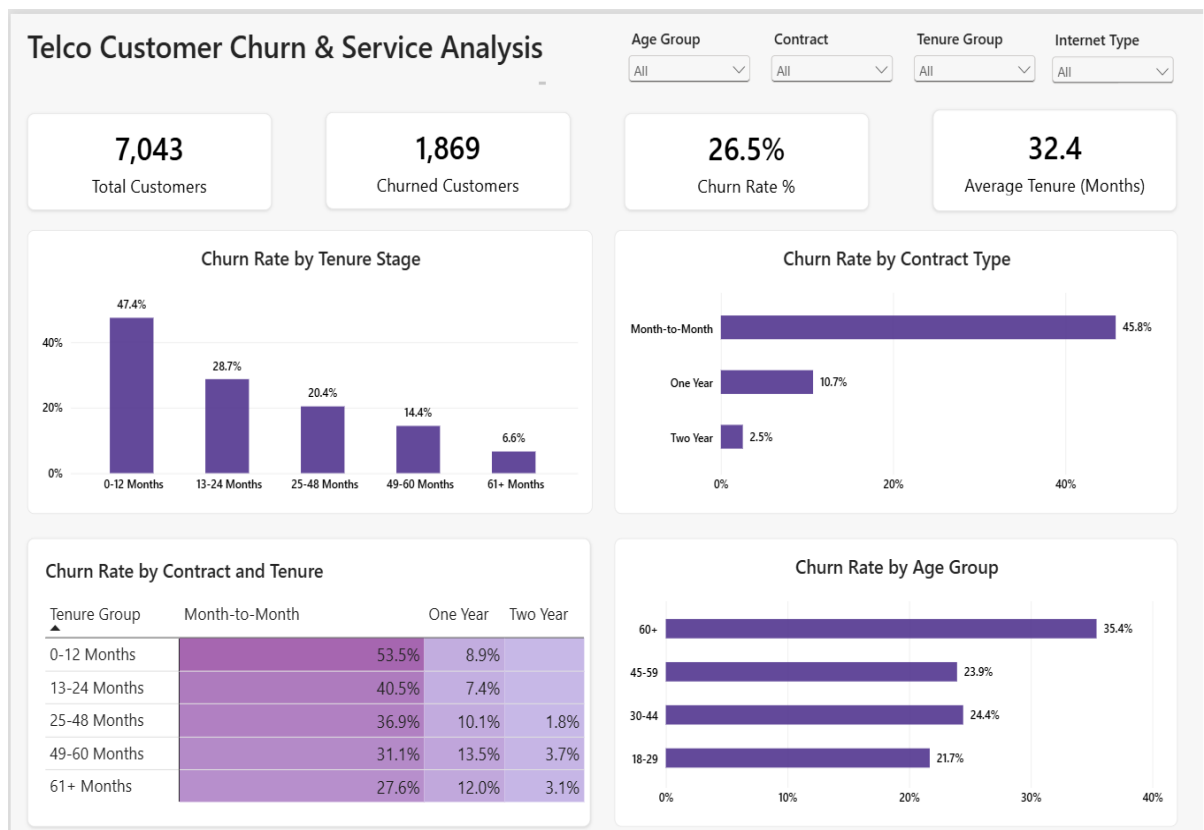


Figure 1 – Power BI Dashboard: Telco Customer Churn & Service Analysis

5. Business Recommendations

Based on the observed churn patterns, the company should:

1. **Prioritize further investigation of early-tenure customers**, particularly during the first 12 months, through stronger onboarding and engagement.
2. **Investigate month-to-month customers**, as this segment shows substantially higher observed churn across tenure stages.
3. **Investigate high-churn service segments**, particularly Fiber Optic customers, to understand potential service or experience-related factors.
4. **Use customer segmentation** based on age, household status, referrals, services, tenure and contract type to identify groups requiring further investigation.

6. Limitations

- The dataset does not explain why customers churn beyond the available customer and service attributes.
- Some service-related fields contain expected missing values.
- The analysis does not measure the financial impact or cost of retention actions.

7. Conclusion

The analysis of 7,043 customers identified the strongest observed churn patterns around early-tenure and month-to-month customers. Churn also varies across age, household characteristics, referral engagement, service adoption and internet service type.

The Power BI dashboard summarizes these patterns and provides a clear basis for identifying customer segments that warrant further investigation.